

A dozen space stations?

Did you know that more than a dozen space stations have been launched between 1971 and 2012? The earliest space stations didn't really last long in orbit though

**A salute to Salyut**

Salyut 1 was the very first space station ever to be launched into space. Due to technical issues, it was functional for only 23 days

The TRUE purpose of the space station

At an estimated cost of over \$35 billion, the International Space Station is one very expensive satellite. What is the true purpose of the ISS?

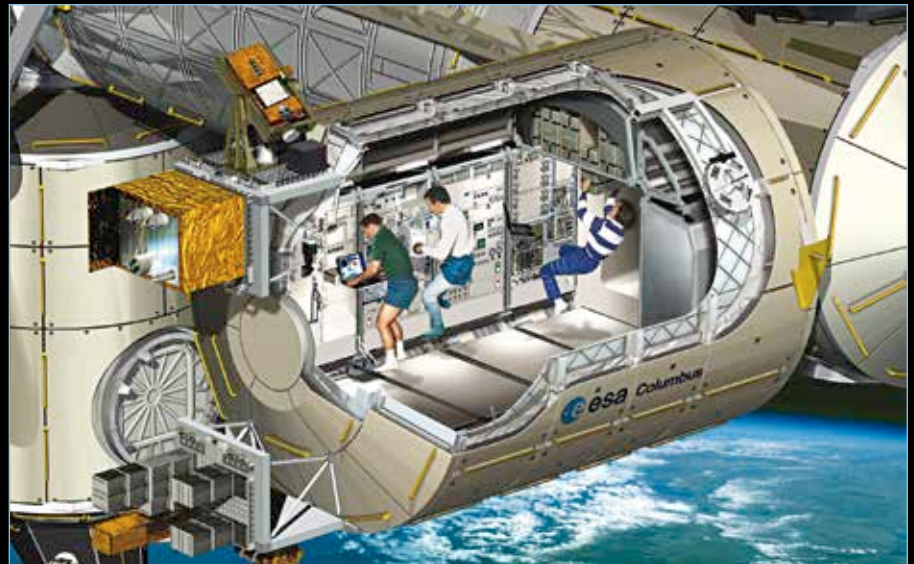
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The current space station, better known as the International Space Station, has been in orbit for over a decade and has swallowed a very large portion of many a country's resources. At \$35 billion dollars (probably much more) at the very least, you have to wonder if it is really worth all that effort, especially as China has now embarked on its mission to setup its own space station, the Tiangong series, one that can rival the current ISS in terms of technology and overall functionality.

A history lesson

The very first functional Space Station that was put into orbit was the Russian Salyut 1. Launched in 1971, this was a very basic spacecraft and while technically a space station, it wasn't really anything more than a crew compartment with nominal research facilities. For the Russians (and the world in general for that matter), this station was the first step in establishing that long-term survival in space was possible. The initial Salyut series were very primitive by today's standards, the Payload of the current Space Shuttles is far more than the weight of the entire station itself.

The Russians were of course, happy to piggy-back their Almaz stations along with the Salyut, the Almaz series being entirely military spacecraft. Meanwhile, the Americans went ahead and launched their Skylab station. This was far more advanced than the Russian stations and had far better facilities and observational tools.



The purpose

The problem however wasn't that the stations could be launched; that was established a long time ago. These primitive stations were far too simple to actually serve much of a functional purpose considering that satellites even at that time would have been far better at conducting the very same experiments that the space stations were deployed to conduct in the first place, and at a fraction of the cost. It was like having a manned Hubble space telescope. The human element served no essential purpose on those stations. In fact, senior scientists in NASA had rejected the initial proposals for the space station programs on the grounds that the stations were redundant and even more so, unnecessary.

The government did eventually go ahead with the plan. What they didn't accept at the time was that the space stations were just, to put it mildly, nothing more than a game of one-upmanship between the Russians and the Americans. There was nothing wrong with this of course, the Russians were quite proud of their stations for what they were, a symbol of their superiority in the "space age". The Americans pretended otherwise but they knew the truth.

MIR and the ISS

These were the first truly international stations. Primarily made with collaboration between the Russians and the US, these stations, while expensive, were proper space laboratories. Despite the cost of maintaining these stations, \$8 per month per American citizen by some estimates, the kind of research being conducted is claimed to be generating as much and more in returns.

Their research into microbiology (astrobiology), astronomy, human research and even Space Weather is revolutionary. Even the technologies used to build and design the space station are some of the most revolutionary the world has ever seen. This research includes fuel cells, development in solar cell technology, entirely new strains of fruits and vegetables, liquid metal... the list goes on. The developments and advances in human conditioning couldn't have been done anywhere else. If we plan to conquer even just our solar system in the near future, without the space stations, these goals would have been set back by decades.

So what is the true cost of the space station? Priceless. 